

CITY UNIVERSITY OF HONG KONG

Distributed Fixed-Time Nash Equilibrium
Seeking Algorithms for Non-Cooperative
Games in Multi-Agent Systems

Submitted to

Department of Mechanical Engineering

in Partial Fulfillment of the Requirements
for the Degree of Doctor of Philosophy

by

Ouyang Binghao

June 2026

Abstract

Game theory, as an approach to decision-making, has been applied widely in numerous fields such as artificial intelligence, robust control, social policymaking in political science and distributed optimization in multi-agent systems. Non-cooperative games are one of the most useful games in modeling and analyzing interactive decision-making processes, in which players are self-interested to optimize their own actions. As a key concept for non-cooperative games, Nash equilibrium, refers to an action profile in which no player can gain more payoff by unilaterally changing its own action and has been widely adopted in analysis of non-cooperative games. Based on whether or not each individual player has direct access to other players' actions, Nash equilibrium seeking approaches can be classified into two categories, that is, centralized algorithms and distributed algorithms. However, centralized algorithms, where the centralized information is to be shared by all players, would run into difficulties for large-scale network-based games. As a result, distributed algorithms, where only partial information is to be shared by neighboring players, have been receiving increasing attention. Note that, in most existing works on distributed Nash equilibrium seeking, the proposed algorithms converge to the Nash equilibrium asymptotically or exponentially. In other words, the Nash equilibrium is reached only when time goes to infinity. More recently, distributed finite-time Nash equilibrium seeking problems have been further investigated due to their faster convergence. However, the upper bound of the convergence time of the finite-time algorithms is dependent on the initial state of the underlying system and thus potentially becomes very large when the initial state deviates far away from the equilibrium. To fill this gap, distributed fixed-time Nash equilibrium seeking algorithms, where the upper bound of the convergence time is independent of the initial conditions of the systems, deserve further investigation.

Furthermore, distributed Nash equilibrium seeking for multi-agent systems in practice often encounters severe challenges, such as vulnerable communication networks under denial-of-service (DoS) attacks, which disrupt information flow and degrade

seeking performance; restricted decision spaces in constrained games, where any boundary violation may lead to system instability; and players governed by heterogeneous linear and nonlinear dynamics, making the design and analysis of the distributed fixed-time algorithms much more challenging. Addressing these challenges is crucial for real-world engineering applications, many of which explicitly call for fixed-time convergence.

Motivated by these observations, this dissertation systematically investigates distributed fixed-time Nash equilibrium seeking problems for multi-agent systems. The main contributions are outlined from four progressive aspects:

Part I: This part investigates the distributed fixed-time Nash equilibrium seeking problem for non-cooperative games. A novel distributed fixed-time Nash equilibrium seeking algorithm with a two time-scale framework is proposed based on a leader-following consensus protocol and the sliding mode theory. Different from existing works on asymptotic, exponential, or finite-time convergence, the proposed algorithm achieves fixed-time convergence with an explicitly given upper bound for the convergence time, which is independent of the initial conditions of the concerned system. Moreover, unlike the coupled passivity-based design in existing works on distributed fixed-time Nash equilibrium seeking, where verification of parameter feasibility involves solving some complex inequalities that can be challenging for large-scale systems, the proposed algorithm with a two time-scale framework can guarantee the existence of the design parameters and brings great benefits for the selection of the design parameters. Furthermore, as a significant extension, the proposed algorithm is modified to address time-varying quadratic games, ensuring fixed-time tracking of the Nash equilibrium trajectory.

Part II: This part addresses the resilient distributed fixed-time Nash equilibrium seeking problem under denial-of-service (DoS) attacks. Since communication networks of multi-agent systems are inherently vulnerable to malicious DoS attacks, which destroy network connectivity and compromise information timeliness, a resilient distributed fixed-time Nash equilibrium seeking algorithm is developed. Based on a leader-following consensus protocol, switching system theory, and sliding mode theory, the proposed algorithm ensures that the actions of the players can still converge to the Nash equilibrium in fixed time despite the presence of DoS attacks, with the upper bound of the convergence time explicitly given. Furthermore, as a notable enhancement for scenarios where DoS attack intervals are known, a time-varying gains scheme is proposed. By dynamically assigning high gains in the absence of attacks and low gains

during DoS intervals, this extended scheme enables the proposed algorithm to tolerate DoS attacks of significantly higher intensity while preserving fixed-time convergence.

Part III: This part explores the distributed Nash equilibrium seeking problem for constrained non-cooperative games. Since a player's feasible action space is often restricted by physical limitations in practice, constrained non-cooperative games have attracted extensive attention. Based on variational inequality theory, the problem of Nash equilibrium seeking for constrained non-cooperative games can be reformulated as a variational inequality problem. Building upon this reformulation, a novel distributed algorithm is developed by combining neurodynamic methods, a leader-following consensus protocol, and sliding mode theory. It is also proven that under the proposed algorithm, the actions of all players converge to the Nash equilibrium in a fixed time, which is independent of the initial conditions of the players and the upper bound of the convergence time for the proposed Nash equilibrium seeking algorithm is explicitly given.

Part IV: This part tackles the distributed fixed-time Nash equilibrium seeking problem for multi-agent systems with heterogeneous and complex physical dynamics. While the majority of existing algorithms are confined to first-order and homogeneous systems, real-world engineering applications often involve players governed by heterogeneous and complex dynamics, such as high-order integrator dynamics, general linear dynamics, and nonlinear Euler-Lagrange dynamics. To handle such heterogeneous dynamics, a novel two-layer design framework consisting of a virtual layer and a physical layer is established. In the virtual layer, distributed fixed-time Nash equilibrium seeking algorithms are utilized to generate a virtual signal that converges to the Nash equilibrium. In the physical layer, fixed-time tracking protocols are specifically designed for the heterogeneous physical dynamics so that their actual actions converge to the virtual signals. It is proven that under this decoupled framework, the physical actions of all heterogeneous players converge to the Nash equilibrium in fixed time regardless of initial conditions. Moreover, the upper bounds for their convergence times can be explicitly given.

Finally, to bridge the gap between theoretical design and practical implementation, the effectiveness and practicality of the proposed distributed algorithms are presented through three distinct engineering scenarios: a distributed energy trading game in smart microgrids, distributed computation offloading in vehicular edge computing systems, and target protection for air-ground heterogeneous multi-agent systems with conflicting goals.

Contents

Abstract	ii
List of figures	ix
List of tables	x
1 Introduction	1
1.1 Background	1
1.2 Literature Review	4
1.2.1 Nash Equilibrium Seeking of Multi-Agent Systems	4
1.2.2 Distributed Nash Equilibrium Seeking Under Denial-of-Service Attacks	6
1.2.3 Distributed Nash Equilibrium Seeking for Constrained Non-Cooperative Games	6
1.2.4 Distributed Nash Equilibrium Seeking for Players of Complex Dynamics	7
1.2.5 Finite-Time/Fixed-Time Distributed Nash Equilibrium Seeking	7
1.3 Motivations for This Thesis	8
1.4 Thesis Contributions	9
1.5 Thesis Organization	12
2 Preliminaries	14

2.1	Graph Notations	14
2.2	Denial-of-Service Attacks	14
2.3	Variational Inequalities and Neurodynamics	15
2.4	Some Mathematical Results	15
2.5	Definitions and Lemmas on Fixed-Time Stability	17
3	Distributed Fixed-Time Nash Equilibrium Seeking for Non-Cooperative Games	19
3.1	Introduction	19
3.2	Problem Formulation	22
3.3	Main Results	23
3.3.1	Distributed Nash Equilibrium Seeking Algorithm	23
3.3.2	Convergence Analysis	24
3.3.3	Extension to Time-Varying Quadratic Games	30
3.4	Numerical Examples	30
3.5	Conclusions	31
4	Resilient Distributed Fixed-Time Nash Equilibrium Seeking Against Denial-of-Service Attacks	33
4.1	Introduction	33
4.2	Problem Formulation	35
4.3	Main Results	37
4.3.1	Resilient Distributed Nash Equilibrium Seeking Algorithm	37
4.3.2	Convergence Analysis	38
4.3.3	Extension to Known DoS Attack Intervals	46
4.4	Numerical Examples	47
4.5	Conclusions	47

5	Distributed Fixed-Time Nash Equilibrium Seeking for Constrained Non-Cooperative Games	49
5.1	Introduction	49
5.2	Problem Formulation	51
5.3	Main Results	53
5.3.1	Distributed Nash Equilibrium Seeking Algorithm	53
5.3.2	Convergence Analysis	55
5.4	Numerical Examples	63
5.5	Conclusions	68
6	Distributed Fixed-Time Nash Equilibrium Seeking for Multi-Agent Systems with Heterogeneous Complex Dynamics	70
6.1	Introduction	70
6.2	Problem Formulation	72
6.3	Main Results	75
6.3.1	Virtual Layer Design for Nash Equilibrium Seeking	76
6.3.2	Physical Layer Control for Heterogeneous Dynamics	76
6.3.3	Convergence Analysis	79
6.4	Numerical Examples	85
6.5	Conclusions	85
7	Engineering Applications and Verification	86
7.1	Introduction	87
7.2	Application I: Distributed Energy Trading Game in Smart Microgrids	87
7.2.1	Background	87
7.2.2	Problem Formulation	87
7.2.3	Algorithm Deployment	87
7.2.4	Simulation Results and Discussion	87

7.3	Application II: Distributed Computation Offloading in Vehicular Edge Computing Systems	87
7.3.1	Background	87
7.3.2	Problem Formulation	88
7.3.3	Algorithm Deployment	90
7.3.4	Simulation Results and Discussion	90
7.4	Application III: Target Protection of Air-Ground Heterogeneous Multi-Agent Systems with Conflicting Goals	90
7.4.1	Background	90
7.4.2	Problem Formulation	90
7.4.3	Algorithm Deployment	90
7.4.4	Simulation Results and Discussion	90
7.5	Conclusions	90
8	Conclusions and Future Work	91
8.1	Conclusions	91
8.2	Future Work	91
	Bibliography	93